

Reliability Database for Wide Bandgap Power Semiconductors and Module Packaging via Comprehensive Survey and Standardization

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Abstract—The reliability of power electronic (PE) systems is becoming increasingly critical as these technologies are integrated into mission-critical applications. Accurate lifetime prediction for high-failure-rate components, such as power semiconductors and packaging, is essential for ensuring long-term efficient system performance. However, developing high-fidelity lifetime models requires a thorough understanding of underlying physics-of-failure (POF) mechanisms and extensive stress testing, which is often prohibitively costly and time-consuming when conducted individually. This study addresses these challenges by establishing a centralized reliability database, developed through a comprehensive survey of failure mode and effects analysis (FMEA) for the most vulnerable components in PE systems. Lifetime models associated with dominant failure mechanisms are compiled and standardized, including the extraction of key model coefficients, to support consistent and efficient reliability assessments. The resulting database serves as a valuable resource for benchmarking and guiding future research. The focus is placed on wide bandgap (WBG) power semiconductors and module packaging, two of the most failure-prone elements in modern power electronic converters. Additionally, this study identifies existing research gaps and proposes a structured roadmap for advancing the development of a comprehensive reliability database.

Index Terms—Failure mechanism, GaN, module packaging, reliability model, SiC, wide bandgap.

I. INTRODUCTION

PE technologies being increasingly deployed in mission-critical applications render reliability assessment essential for evaluating operational lifetimes and underlying failure mechanisms. Typically, approximately 90% of faults are attributed to high-failure rate components such as capacitors, semiconductors, solder joints, and printed circuit boards (PCBs) [1]. Among the predominant stressors contributing to component degradation, temperature cycling, humidity and moisture exposure, and mechanical vibrations or shocks account for nearly 96% of failures in systems like solar inverters [1]. A conventional approach to reliability evaluation involves accelerated stress testing, where the components undergo prolonged aging under elevated stress conditions. While effective and accurate, this method is resource-intensive in terms of time, cost, and labor [2], [3]. To circumvent these limitations, this study proposes the development of a unified reliability database. This involves conducting a comprehensive literature survey to identify the most failure-prone PE components, summarizing their

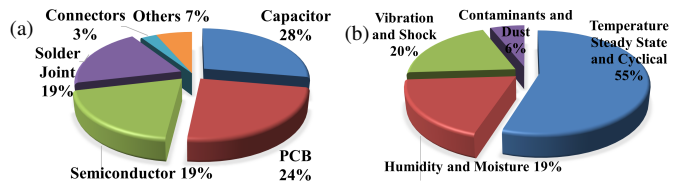


Fig. 1. (a) Major fault locations in a power electronic (PE) converter, and (b) critical stressors leading to degradation of high-failure rate components; literature survey is conducted based on solar application [1].

degradation mechanisms, collecting the corresponding lifetime models, and standardizing respective model formats and coefficients. The resulting database aims to serve researchers and practitioners across the PE community by providing validated models for efficient reliability prediction.

This paper focuses specifically on the WBG devices and module packaging, two emerging and mission-critical PE technologies undergoing rapid development. Although significant research exists on the reliability of individual WBG devices and packaging structures, these efforts remain fragmented. Thus, a systematic and collaborative approach to reliability modeling and determination/estimation of the model-based coefficients is needed. To that end, this study integrates previous research into a structured framework for understanding degradation in WBG devices and packaging. Key failure locations, including semiconductor devices, wire-bonds, solder joints, and the corresponding lifetime models, are compiled based on POF mechanisms. Critical model coefficients are extracted to facilitate standardized lifetime prediction. These contributions not only streamline future reliability assessments but also offer valuable benchmarks for the community, reducing dependence on costly traditional testing methods. Furthermore, this study also aims to highlight knowledge gaps and propose a roadmap for future research, thus supporting the development of more robust and reliable power electronic systems.

Remainder of this paper is organized as follows. Section II covers Gallium Nitride (GaN) power semiconductors, Section III addresses Silicon Carbide (SiC) power devices, and Section IV discusses module packaging. Each section begins

II. RELIABILITY STUDY FOR GAN DEVICES

A. Failure Locations and Mechanisms in GaN Devices

2) *p-GaN*: Time-dependent breakdown in the p-GaN region due to defect percolation is reported in case of HEMTs [7]–[9]. This is commonly modeled using power-law and exponential formulations with Arrhenius-type temperature dependence.

3) *2DEG Channel*: The key failure mechanisms are:

b) Hot Carriers Trapping: During switching of GaN HEMTs, electrons can gain sufficient energy to overcome the AlGaIn barrier and escape the 2DEG channel. These electrons may become trapped in various regions of the device, including passivation, buffer, GaN bulk, AlGaIn/passivation interface, eventually leading to increased in turn-on resistance. This mechanism has been modeled using exponential relations, some incorporating additional power-law or logarithmic terms in the exponent—derived from underlying physical processes [11]–[13]. For temperature dependence, classical Arrhenius relationship is reported, while an additional square-root-of-temperature term is used in [11].

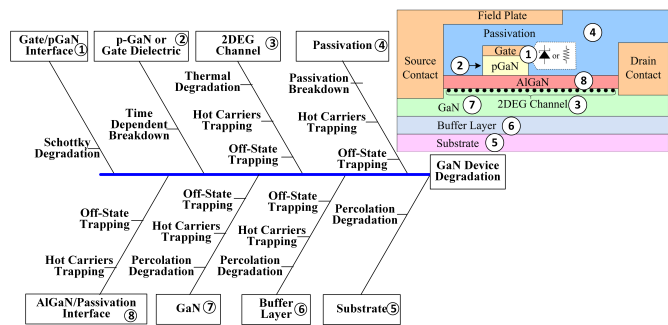


Fig. 2. Fish bone diagram depicting various failure locations congruent with different failure mechanisms for GaN devices. A p-GaN HEMT, the most commonly available commercial GaN HEMT, are shown at top right.

4) *Passivation Layer:* Under a positive gate bias, electrons traveling through the p-GaN layer generate additional electron-hole pairs due to impact ionization. This results in holes being trapped in the passivation dielectric, ultimately leading to passivation breakdown. Models often combine exponential and power-law terms with linear or physics-based temperature relationships as reported in [11], [15].

5) *GaN Bulk*: During reverse biasing, percolation-induced degradation in the GaN bulk has been reported to cause increased leakage. For gate injection transistor (GIT), this is modeled using an exponential model with an Arrhenius temperature relationship [16].

6) *Substrate*: Similarly, percolation in the substrate layer contributes to vertical leakage percolation degradation [16].

B. Summary and Overview

P-type Schottky gate contacts are most common among the commercialized GaN HEMTs, with lifetime models mainly focusing on Schottky degradation modeled using power and exponential terms. An Arrhenius temperature relationship with negative activation energy of -0.3 eV is reported in [4], [6]. Similarly, an exponential temperature relation up to 100 °C was proposed in [5]. Passivation breakdown is modeled using a power-law term (exponent of 1.9) [11], though temperature dependence varies. Time-dependent breakdown in p-GaN gate structures follows power laws, with activation energies of $0.1 - 0.9$ eV [17], [18]. Thermal stress in the 2DEG channel is modeled with an Arrhenius relation and activation energy of 1.13 eV [10]. Hot carrier trapping and off-state trapping are modeled using exponential laws with activation energies of $0.09 - 0.1$ eV and 3.22 eV, respectively [11], [13], [14]. Percolation degradation has an activation energy of 0.62 eV [16], and power-law models without specific failure mechanisms report activation energies $0.03 - 0.71$ eV [19], [20].

TABLE I
SUMMARY FOR STATE OF THE ART IN GAN DEVICE DEGRADATION

Publication	Failure mechanism	Critical stressors	Failure criterion	Model format	Model parameters ^a
Wang (2024) [4]	Time dependent Schottky breakdown	Gate voltage, Temperature	Sudden increase in the gate current	$\mathcal{TTF} = A(V_g - V_0)^{-n_v} f_{sw}^{-n_f} e^{\frac{Q_a}{k_B T}}$	$V_0 = 4 \text{ V}, n_v = 26, Q_a = -0.3, n_f = 0.6$ ^b
He (2019) [6]				$\mathcal{TTF} = AV_g^{-n_v} e^{-\beta_T T}$	$\beta_T = -ve \forall T < 100 \text{ }^\circ\text{C}, = 0 \forall T > 100 \text{ }^\circ\text{C}$ ^c
Greco (2024) [5]		Gate current density		$\mathcal{TTF} = Ae^{\beta_i I_g^{-\gamma_i}} e^{\frac{Q_a}{k_B T}}$	$\gamma_i = 0.5$ ^d
Pozo (2022) [11]	Passivation breakdown	Gate voltage, Temperature	Not specified	$\mathcal{TTF} = A(A_T T + 0.35) e^{\beta_v (V_g + V_0) - \gamma_v} =$	$A = 1.7 \cdot 10^{-6}, A_T = 1.7 \cdot 10^{-6}, \beta_v = 2.17 \cdot 10^{-3}, \gamma_v = 1.9, e$
Gao (2024) [15]		Electric field, Temperature	Sudden increase in the gate leakage current	$\mathcal{TTF} = Ae^{\beta_e E^{-\gamma_e}}$	$\beta_e = (1.07 \cdot 10^7 + 6.34 \cdot 10^3 T)^{\gamma_e}, \gamma_e = 1.9$ ^f
Tallarico (2017) [7]	Time-dependent p-GaN breakdown	Gate voltage, Temperature	Gate current exceeding 1 mA/mm	$\mathcal{TTF} = A(1 + e^{\frac{\beta_i}{I_g}}) e^{\frac{Q_a}{k_B T}}$	$Q_a = 0.44$ ^g
Tapajna (2016) [8]		Gate current, Temperature	Sudden increase in the gate current and noise	$\mathcal{TTF} = AI_g^{-n_i} e^{\frac{Q_a}{k_B T}}$	$Q_a = 0.1$ ^h
Veerreddy (2017) [9]		Drain-source voltage, Temperature		$\mathcal{TTF} = AV_{ds}^{-n_v} e^{\frac{Q_a}{k_B T}}$	$n_v = 22.8, Q_a = 0.9$ (gr. I) $n_v = 28.1, Q_a = 0.71$ (gr. II) ⁱ
He (2020) [17]	Generic gate lifetime models	Gate voltage	Not specified	$\mathcal{TTF} = Ae^{\beta_v V_g^{-\gamma_v}}$	$\beta_v = +ve, \gamma_v = +ve$ (cond. 1), $\beta_v = -ve, \gamma_v = -0.5$, (cond. 2), $\beta_v = +ve, \gamma_v = 1$, (cond. 3) ^j
Stockman (2018) [18]		Gate current	Not specified	$\mathcal{TTF} = AI_g^{-\gamma_i}$	$\gamma_i = 0.25$ (gr. I), $n_i = 1$ (gr. II) ^k
Sayed (2023) [10]	Thermal stress at 2DEG layer	Temperature	25% increase in on resistance	$\mathcal{TTF} = Ae^{\frac{Q_a}{k_B T}}$	$Q_a = 1.13$ ^l
Pozo (2022) [11]	Hot carriers trapping	Drain-source voltage, Temperature	20% increase in on resistance	$\mathcal{TTF} = A\sqrt{T} e^{\frac{Q_a}{k_B T}} e^{-\beta \log(1 + e^{n_v(V_{ds} - V_{fd})})} =$	$Q_a = 0.092$ ^m
Yin (2022) [12]			30% increase in on resistance	$\mathcal{TTF} = Ae^{-\beta_v V_{ds}} e^{\frac{Q_a}{k_B T}}$	$\beta_v = 0.03$ ⁿ
Ikoshi (2018) [13]		Hard-switching, Temperature	Not specified	$\mathcal{TTF} = Ae^{-\beta_v V_{ds}} e^{-\beta_i I_{ds}} e^{\frac{Q_a}{k_B T}}$	$\beta_v = 0.039, \beta_i = 0.47, Q_a = 0.1$ ^o
Ma (2024) [14]	Off-state trapping	Drain-source voltage, Temperature	Not specified	$\mathcal{TTF} = AV_{ds}^{-n_v} e^{\frac{Q_a}{k_B T}}$	$n_v = 24.45, Q_a = 3.22$ ^p
Tanaka (2017) [16]	Percolation degradation	Drain-source voltage	Leakage current > 15 μA	$\mathcal{TTF} = Ae^{-\beta_v V_{ds}} e^{\frac{Q_a}{k_B T}}$	$\beta_v = 0.037, Q_a = 0.62$ ^q
Gandhi (2019) [19]	Generic device lifetime models	Drain-source voltage, Temperature	Not specified	$\mathcal{TTF} = AV_{ds}^{-n_v} e^{\frac{Q_a}{k_B T}}$	$n_v = 17.2, Q_a = 0.71$ ^r
Wang (2022) [20]			15% increase in the gate leakage current		$Q_a = 0.0336$ (gr. I), $Q_a = 0.303$ (gr. II) ^s

^a k_B is the Boltzmann constant = 8.617×10^{-5} eV/K; Q_a is the activation energy in eV; E is electric field in V/cm; A is empirical constant in s; $n_f, n_i, n_v, \beta_i, \beta_v, \gamma_i$ and γ_v are model based exponents derived from experimental results based on commercial/custom-built devices.
^b device under test (DUT) is a commercial 650 V, 30 A Schottky contact p-GaN (SP-GaN) HEMT [4]; n_f is derived $\forall 100 \text{ kHz} < f_{sw} < 3.1 \text{ MHz}$.
^c DUT is commercial 650 V, 200 m Ω SP-GaN HEMTs [6].
^d DUT is a custom-built 650 V, 200 m Ω SP-GaN HEMT [5].
^e DUT is EPC 2212 [11].
^f DUT is EPC 2252 [15].
^g DUT is custom-fabricated SP-GaN HEMT with $V_{th} = 2.1 \text{ V}$ [7].
^h DUT is custom-fabricated SP-GaN HEMT [8].
ⁱ DUTs are custom-built 600 V GaN HEMTs, where group II device is fabricated using different gate process reducing drain leakage current [9].

^j DUT is commercial 650 V 500 m Ω SP-GaN HEMT [17]; exponent β_v is derived based on condition 1: $-25 \text{ }^\circ\text{C} < T < 50 \text{ }^\circ\text{C}$, condition 2: $T < -25 \text{ }^\circ\text{C}, V_g < 10 \text{ V}$, and condition 3: $T < -25 \text{ }^\circ\text{C}, V_g > 10 \text{ V}$.
^k DUT under group-I has non-self aligned gate metal whereas device under group-II has additional layer between gate stack and p-GaN [18].
^l DUT is 650 V, 7.5 A SP-GaN HEMT with $R_{ds(on)} = 180 \text{ m}\Omega$ [10].
^m DUT is EPC2045 claimed applicable to 100 V EPC Gen-5 products [11].
ⁿ DUT is 650 V 105 m Ω GaN device with strain enhancement layer [12].
^o DUT is 600 V 10 m Ω HDGIT with DFN package and $V_{th} = 1.2 \text{ V}$ [13].
^p DUT is 650 V 11 A SP-GaN GaN (INN650D02) [14].
^q DUT is 600 V 70 m Ω hybrid drain gate injection transistor (HDGIT) with dual flat no-lead (DFN) package and threshold voltage $V_{th} = 1.2 \text{ V}$ [16].
^r DUT is Navitas device [19].
^s DUTs under group I and II have gate width of 3.6 mm and 12.5 mm [20].

Being relatively newer, various GaN HEMT manufacturers employ very different device structures and their reliability is expected to be different. Additionally, failure mechanisms are not yet fully understood. This is reflected in the lack of common formats across different surveyed literature. Despite significant efforts in model development, few have provided lifetime parameters. Additionally, the complexity of some physics-based models, along with the use of non-measurable parameters like electric field and current density, makes it challenging to fit and calculate their lifetime accurately.

III. RELIABILITY STUDY FOR SiC DEVICES

Electrical stress can lead to chip-level degradation of Silicon Carbide (SiC) devices, including gate oxide wear-out, bipolar degradation, and single-event burnout. Given these reliability challenges, this section provides a concise overview of the primary failure locations and mechanisms affecting SiC devices, with a particular emphasis on die-level degradation. Additionally, it introduces a reliability model with relevant coefficients that can be utilized in power electronics applications to estimate the device lifetime or mean time to failure (MTTF), thereby aiding in the design of more robust and failure-resistant power systems. These mechanisms and their associated failure locations are visualized using fishbone diagram shown in Fig. 3. Furthermore, the corresponding reliability models and coefficients are summarized in Table II.

A. Failure Locations and Mechanisms in SiC Devices

1) *Gate Oxide Region*: Gate oxide reliability in SiC MOSFETs is challenged by intrinsic material properties and high-field operation. SiC's wide bandgap (3.2 eV) and lower SiC/SiO₂ barrier height promote electron tunneling, increasing leakage and accelerating oxide degradation [21]. Higher operating fields further exacerbate time-dependent dielectric breakdown (TDDB), especially at trench corners and gate edges due to field crowding [22]. Additionally, SiC oxidation yields a high density of interface and near-interface oxide traps, leading to threshold voltage instability, mobility degradation, and premature dielectric failure [23].

a) *Time-Dependent Dielectric Breakdown (TDDB)*: This is a key reliability concern in SiC MOSFETs, resulting from the gradual degradation of the SiO₂ gate oxide under prolonged electrical stress. Defects such as broken Si–O bonds, oxygen vacancies, and trapped charges accumulate within the oxide over time [24], [25]. When defect density reaches a critical threshold, a conductive percolation path forms between the gate and channel, leading to significant current flow, localized Joule heating, and ultimately catastrophic oxide failure [26].

TDDB behavior is typically described by two models. The E-model, dominant at moderate fields and elevated temperatures, attributes breakdown to field-assisted weakening of Si–O bonds and thermally activated defect generation, with time-to-failure showing exponential dependence on the electric field [27]. In contrast, the 1/E-model, relevant under high-field conditions, involves electron injection via Fowler–Nordheim tunneling and impact ionization within the oxide. Trapped

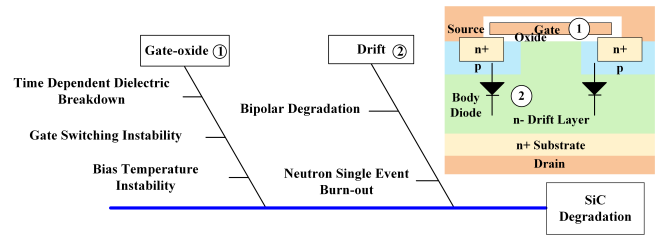


Fig. 3. Fish bone diagram showing different failure mechanisms associated with major failure locations in case of SiC device.

holes near the SiC/SiO₂ interface enhance local electric fields, accelerating defect accumulation and breakdown [28]–[30].

b) *Bias Temperature Instability (BTI)*: This failure mechanism, particularly positive bias temperature instability (PBTI), is a key reliability issue in nase of N-channel SiC MOSFETs under positive gate bias. It manifests as a threshold voltage shift, degrading device performance [31], [32]. This degradation is linked to various oxide-related charges, including interface traps (N_{it}), near-interface oxide charges (N_{ot}), fixed charges (N_F), and mobile ions (N_M), which interact differently with the applied electric field [21], [33], [34]. Shift in threshold voltage primarily result from the charging/discharging of N_{it} and N_{ot} , as well as stress-induced trap generation [35]. While interface trap effects are typically fast and reversible, near-interface oxide traps evolve more slowly and are less recoverable, thereby contributing to long-term reliability degradation in SiC MOSFETs.

c) *Gate Switching Instability (GSI)*: This refers to the threshold voltage shift (ΔV_{th}) in SiC MOSFETs caused by high-frequency gate switching between positive and negative voltages [36], [37]. Initially misidentified as AC bias temperature instability, GSI is now recognized as a distinct mechanism driven by cumulative switching cycles, particularly under negative gate bias [38], [39]. Unlike BTI, GSI shows negligible dependence on temperature or positive gate bias. GSI occurs only when the gate swings between inversion (positive bias) and accumulation (negative bias), and not when switching between positive and zero bias [37]. The total threshold voltage shift is the sum of BTI and GSI effects: BTI dominates at low cycle counts, while GSI dominates at higher switching cycles, and is mainly affected by negative gate bias [37].

B. Summary and Overview

The key failure mechanisms in MOSFETs include gate oxide degradation, such as TDDB, negative and positive bias temperature instabilities (NBTI/PBTI), and package degradation, such as bond wire failure, solder degradation, etc. TDDB reliability is modeled using the Arrhenius equation, where the time to failure depends on temperature—with an activation energy around 1 eV—and electric field, with field acceleration factors ranging from 1.15–4.33 cm/MV (for $E < 8.5$ MV/cm) to 4.6–8.27 cm/MV (for $E > 8.5$ MV/cm) [24], [28], [29], [40]–[44]. The electric field around $E \approx 8$ –8.5 MV/cm represents the transition region above which degradation is accelerated by impact ionization and anode hole injection. A

TABLE II
SUMMARY FOR STATE OF THE ART IN GAN DEVICE DEGRADATION

Publication	Failure mechanism	Critical stressors	Failure criterion	Model format	Model parameters ^a
Yu (2010) [24]	TDDDB (gate oxide degradation)	Electric field, Temperature	Increased gate leakage current	$MTTF = Ae^{\left(-\gamma_e E_{ox} + \frac{Q_a}{k_B T}\right)}$	$\gamma_e = 3.45 \forall E < 9,$ $Q_a = 0.9 \forall 8.09 < E < 9.36$ ^b
Le-Huu (2011) [40]					$\gamma_e = 2.53 - 2.901 \forall 7 < E < 9.5,$ $Q_a = 0.77 - 0.79$ ^c
Matocha (2008) [41]					$\gamma_e = 1.15 - 2.30 \forall E < 8.5,$ $\gamma_e = 4.60 \forall E > 8.5$ ^d
Lichtenwalner (2018) [42]					$\gamma_e = 3.53 \forall E < 8.5, Q_a = N/P$ ^e
Liu (2021) [29]					$\gamma_e = 3.54 - 4.33 \forall E < 8.5,$ $\gamma_e = 6.41 - 7.76 \forall E > 8.5, Q_a = N/P$ ^f
Gurfinkel (2008) [43]					$\gamma_e = 2.07 - 3.22 \forall E < 8.5,$ $\gamma_e = 4.6 \forall E > 8.5, Q_a = 1$ ^g
Santini (2015) [44]					$\gamma_e = 8.26 \forall E > 8.5, Q_a = N/P$ ^h
A-Salvadó (2024) [28]				$MTTF = t_0 e^{(\chi_0 - \gamma V_{gs})}$	$\chi_0 = \chi'_0 - (1 - \varphi) \gamma' V_{gs, crit},$ $\varphi = 2.07, V_{gs, crit} = 42.5,$ $\gamma = 1.03, \gamma' = 2.14$ ⁱ
Berens (2020) [45]	NBTI (gate oxide and SiO ₂ /SiC interface)	Electric field, Temperature, Stress time	Change in the threshold voltage	$\Delta V_{th} = AV_{gs}^m t_{stress}^n e^{-\frac{Q_a}{k_B T}}$	$m = 0.08 - 0.19, Q_a = N/P$ ^j
Berens (2020) [45]	PBTI (gate oxide and SiO ₂ /SiC interface)				$n = 0.23 - 0.27, m = 1.58 - 2.46,$ $Q_a = 0.11 - 0.15$ ^l
Volosov (2024) [35]					$m = 8.14, n = 0.27, Q_a = 80 \cdot 10^{-3}$ ^k
Garcia-Mere(2024) [46]	GSI	Number of switching cycles, electric field (negative gate bias), Temperature	Change in the threshold voltage	$\Delta V_{th} = A \left[a \left(\frac{N_c}{b} \right)^{\mathcal{X}} \right]$	$A = 7.94 \cdot 10^{-15},$ $\mathcal{X} = 2.35 \cdot \log(N_c) - 0.096$ $a = -0.18 \cdot V_{neg} - 0.44$ $b = 0.033343 \cdot V_{neg}^2 + 0.468 \cdot V_{neg} + 2.604$ ^m

^a k_B is the Boltzmann constant = 8.617×10^{-5} eV/K; Q_a is the activation energy in eV; E is electric field in MV/cm; V_{gs} is the gate voltage in V; A is empirical constant in s; γ_e is model based exponent expressed in cm/mV, whereas the exponents γ and γ' are shown in V^{-1} .
^b DUT is SiC metal-oxide semiconductor (MOS) capacitor having gate oxide thickness of $t_{ox} = 47$ nm, and tested at $T = 275$ °C [24].
^c DUT is SiC MOS capacitor having gate oxide thickness of $t_{ox} = 28.5$ nm, and tested at temperature range of 200 °C–400 °C [40].
^d DUTs are SiC MOS capacitor and dual-implanted metal-oxide semiconductor field-effect transistor (DMOSFET) having $t_{ox} = 50$ nm and tested at temperature of 200 °C; Q_a is not provided (N/P) in [41].
^e DUT is 1.2 kV, 80 mΩ discrete SiC MOSFET (C2M0080120D) having $t_{ox} = 53$ nm and tested at $T = 150$ °C [42].
^f DUTs are 1.2 kV SiC DMOSFETs having t_{ox} in the range of 41–52 nm, and tested at $T = 150$ °C [29].

^g DUTs are MOS capacitor and DMOSFET having $t_{ox} = 43$ nm, and tested at temperature range of 230 °C–365 °C [43].
^h DUT is 1.2 kV, 24 A SiC DMOSFET having $t_{ox} = 63$ nm, and tested at $T = 150$ °C; failure criterion is gate leakage current $I_{g, lk} = 10$ mA [44].
ⁱ DUT is 1.2 kV SiC MOSFET having $t_{ox} = 53$ nm and tested at 150 °C; the critical gate-source voltage $V_{gs, crit}$ (= 42.5 V) is equivalent to an electric field of $E_{crit} = 8$ MV/cm; thus, exponents γ and γ' can be translated into equivalent values of 5.49 cm/MV and 11.34 cm/MV, respectively [28].
^j DUT is tested at 200 °C and $V_{gs} = -25$ V [45].
^k DUT is 650 V DMOSFET from Microelectronics, and tested at 150 °C, $V_{gs} = 42.5$ V and $t_{stress} = 10^3$ s [35].
^l DUT is tested at 200 °C, $V_{gs} = 25$ V and $t_{stress} = 10^3$ s [45].
^m DUT is 1200 V/117 SiC MOSFET based on trench technology and tested for $V_{neg} = -3$ V to -8 V; a and b depend on V_{neg} whereas X is derived based on number of cycles determined as $N_c = t_{stress} \times f_{sw}$ [46].

change in the threshold voltage characterizes BTI and GSI and is typically expressed in terms of voltage shift rather than MTTF, as it represents degradation rather than catastrophic failure. This threshold voltage shift is modeled using a power-law dependence on V_{gs} and time, along with an Arrhenius-type temperature dependence. The key parameters and relevant literature are summarized in Table II.

In this study, bipolar degradation and neutron-based single event burnout (SEB) mechanisms are excluded. The bipolar mechanism in SiC MOSFETs is closely associated with the operation of the body diode, which conducts during the dead time when both complementary MOSFETs in a phase

leg are turned off. This degradation primarily stems from the recombination-enhanced dislocation glide (REDG) mechanism, triggered by electron-hole recombination during diode conduction. This process compromises device reliability by increasing the forward voltage drop and on-resistance [57]. Since bipolar degradation typically occurs in the early phase of the device lifecycle, corresponding to the infant mortality region of the Weibull distribution, it is classified as an extrinsic failure mechanism. Unlike gate oxide degradation, which is a wear-out phenomenon, bipolar degradation does not follow a predictable aging process; consequently, no established lifetime or MTTF model has been reported for this type of degra-

TABLE III
SUMMARY FOR STATE OF THE ART IN MODULE PACKAGING DEGRADATION

Publication	Failure mechanism	Critical stressors	Failure criterion	Model format	Model parameters ^a
Cui (2005) [47]	Thermo-mechanical stress	Temperature swing, maximum junction temperature	Not specified	$\mathcal{N}_f = A f^m (\Delta T)^{-n} e^{\frac{Q_a}{k_B T}}$	$A = 640, m = 0.33, n = 1.9, f = 22, 40, 16, Q_a = 0.42$ ^b
Held (1997) [48]	CTE mismatch, reconstruction of chip metallization		5% increment in V_{CE} -static measurement	$\mathcal{N}_f = A (\Delta T)^{-n} e^{\frac{Q_a}{k_B T}}$	$A = 640, n = 5, Q_a = 0.8$ ^c
Bayerer (2008) [49]				$\mathcal{N}_f = A (\Delta T)^{-n} t_{on}^{\beta_t} I^{\beta_i} e^{\frac{Q_a}{k_B T}}$	$n = 3.22 - 3.746, \beta_t = -0.438, \beta_i = -0.717, Q_a = 0.2 - 0.129$ ^d
Hernes (2021) [50]			5% increment in V_{CE} , 20% increase in thermal resistance	$A = 5.23 \cdot 10^{15}, n = 4.42, \beta_t = -0.46, \beta_i = -0.72, Q_a = 0.8$ ^e	
Wintrich (2021) [51]			5% increment in V_{CE} -static measurement	$\mathcal{N}_f = A \cdot A_1 (\Delta T)^{-n} t_{on}^{\beta_t} e^{\frac{Q_a}{k_B T}}$	$A = 2.9 \cdot 10^9, n = 7.54 - 4.39, Q_a = 0.28$ ^f
Kostandyan (2012) [52]	CTE mismatch	Temperature swing, mean junction temperature	20% shrinkage of solder inter-connect area	$\Delta \varepsilon_p = A \cdot \Delta T^\beta e^{\frac{Q_a}{RT}}$	$Q_a = 0.3978$ ^g
Baba (2021) [53]		Temperature swing	Change in on-state resistance	$\mathcal{N}_f = A \Delta T^\beta e^{\frac{Q_a}{k_B T}}$	$\beta = -5.96, Q_a = 0.07$ ^h
Gao (2024) [54]				$\mathcal{N}_f = A \Delta T^\beta$	$\beta = -8.8, A = 4.2 \cdot 10^{23}$ ⁱ
Cimmino (2020) [55]	Dendrites and corrosion	Voltage, temperature, humidity bias	End of life	$\mathcal{LF} = A (RH)^{-b} V^{-n} e^{\frac{Q_a}{k_B T}}$	$b = 2.66 - 4.64, n = 2.2, Q_a = 0.79 - 0.95$ ^j
Zorn (2015) [56]					$b = 2.66 - 4.64, n = 2.2, Q_a = 0.79 - 0.95$ ^k

^a k_B is the Boltzmann constant = 8.617×10^{-5} eV/K; α_{Cu} and α_{Si} are the coefficient of temperature expansion (CTE) for copper and silicon, respectively; Q_a is the activation energy in [eV]; n ; A is empirical constant in s; f is the cycling frequency (cycles per day); m is the cycling frequency exponent (typical value 0.33); n is the temperature range exponent, β_i and β_t are model based exponents derived from experimental results.

^b DUT is 6mm×8mm electronic packaging module with three die [47].

^c DUT is 300A/1200V single switch IGBT module [48].

^d DUT is 1200V-IGBT4 module with $I = 10$ A, and $t_{on} = 1.5 - 10$ s [49].

^e DUT is 3300 V/1000 A IHV-type IGBT module having $I = 890$ and 1250 A, and $t_{on} = 0.15 - 4.5$ s [50].

^f DUTs are SiC and IGBT modules from Semikron Danfoss. The constant A_1 depends on chip thickness, time exponent and time coefficient, Coffin-Manson exponent, drop constant, and factor of low temperature change [51].

^g DUT is not mentioned except the wind turbine IGBT module [52].

^h DUT has SOT-227B housing and tested at $\Delta T = 90^\circ\text{C} - 106^\circ\text{C}$, maximum $T_j = 126^\circ\text{C} - 155^\circ\text{C}$, and power loss between 126 W-175 W [53].

ⁱ Based on PCT results [54].

^j DUT is 1200 V power modules with MTP and TO247 packaging [55].

^k DUT is 1.7 kV IGBT module [56].

gradation, and therefore excluded from Table II. On the other hand, SEB is a critical reliability concern for SiC MOSFETs exposed to radiation, often leading to permanent short circuits from energetic particle strikes [42], [58]. Terrestrial neutrons, especially at aviation altitudes, and cosmic heavy ions in space are key sources of SEB risk [59]. SEB in SiC MOSFETs is typically classified as a stochastic failure mode, exhibiting a constant failure rate during standard operational life. This characteristic corresponds to the flat region of the reliability bathtub curve. Therefore, SEB not being a wear-out or aging phenomenon is also excluded from Table II.

IV. RELIABILITY STUDY FOR MODULE PACKAGING

Module packaging in power electronic systems is prone to multiple degradation mechanisms, as evidenced by numerous studies [47]–[56]. Prominent among these are wire bond degradation [48], [49], [53]–[56], solder joint fatigue [50]–[52], and substrate deterioration [47]. These failures are primarily driven by thermo-mechanical fatigue resulting from cyclic thermal loading. Additionally, degradation of the metallization layer

has been reported [48]–[51], largely attributed to junction temperature fluctuations and humidity-induced electrical bias. A comprehensive reliability modeling framework for these packaging-related degradation mechanisms corresponding to the failure mechanisms is illustrated in Fig. 4. The upper section of the diagram captures failure modes predominantly associated with thermo-mechanical fatigue, while the lower section highlights mechanisms influenced by temperature and humidity bias. Table III summarizes the various reliability models associated with different failure mechanism along with their exponents reported in the literature.

A. Failure Locations and Mechanisms in Module Packaging

1) *Bond Wire*: Its degradation is a prevalent failure mechanism in power electronic packaging, resulting from thermo-mechanical stress that leads to crack initiation and propagation at bond-wires [48], [49]. These cracks eventually cause wire bond lift-off, compromising electrical connectivity. The key stressors are junction temperature swing and maximum junction temperature. The time to failure is typically modeled

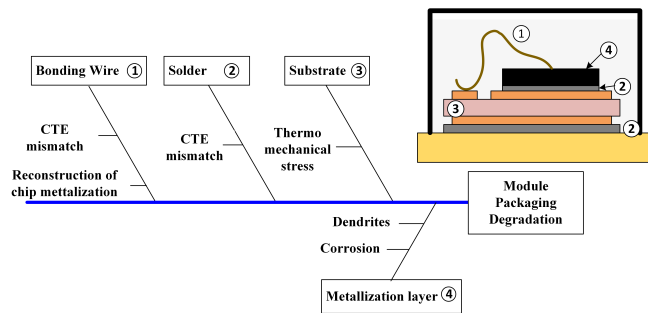


Fig. 4. Fish bone diagram illustrating different failure mechanisms prevalent in various failure locations in the case of module packaging.

using a modified Coffin-Manson relation, which incorporates both inverse power law for junction temperature swing and Arrhenius dependence on the maximum junction temperature to reflect the activation energy of the degradation process.

2) *Solder Layer between Substrate and DBC*: Solder fatigue is induced by thermal cycling due to the mismatch in the coefficients of thermal expansion (CTE) between solder materials and intermetallic compounds [51], [52]. This stress leads to crack initiation at the solder-intermetallic interface, often accelerated by micro-voids within the joint. Cracking impairs mechanical integrity and heat dissipation. The modified Coffin-Manson equation is used to relate the number of cycles to failure with the amplitude of temperature swing and the geometry of solder joint. However, degradation becomes less sensitive to smaller temperature ranges (e.g., below 100 °C).

3) *Substrate Layer*: Substrate degradation primarily results from mechanical fatigue driven by repeated thermal cycling. This leads to cracking or delamination within the substrate structure. A modified Coffin-Manson model, including a frequency-dependent term, is typically used to account for the number of thermal cycles experienced by the component [47].

4) *Metallization Layer*: Metallization layer degradation is commonly associated with electrochemical corrosion occurring in passivated oxide environments. Structural heterogeneities and external bias accelerate metal dissolution via hydrolysis, creating localized pH gradients [55], [56]. This process is further exacerbated by ionic contaminants, which promote electrochemical migration. To model this degradation, an extended form of Peck's model is employed, incorporating a power-law dependence on voltage to capture the impact of temperature, humidity, and electrical stress.

B. Summary and Overview

Similarities across various studies are observed in the coefficients related to junction temperature swing ΔT and activation energy Q_a . Notably, activation energy values exhibit a relatively consistent range across multiple sources. For instance, Q_a values close to 0.80 eV are reported in [48], [50], while other studies indicate values such as 0.42 eV [47], 0.40 eV [52], and a broader range of 0.13–0.80 eV [51] for Si IGBT modules. These findings suggest that thermally activated degradation mechanisms are a common factor, although the

exact activation energies depend on specific materials and operational conditions. The model exponent n also varies, but generally falls within the range of 2 to 7.54, aligning well with the modeling scope presented in this study. Although most of the data are derived from studies on Si IGBTs, it is critical to recognize the growing relevance of WBG semiconductor technologies, therefore, for SiC modules, the packaging degradation was also taken into consideration [53], [54]. It was seen to have 'n' values within the previously mentioned range, although activation energy was seen to be lower but requires further study.

V. CONCLUSIONS

This paper presents a comprehensive overview of the key failure mechanisms in WBG power semiconductors and module packaging, with a focus on associated lifetime and degradation models. Common modeling formats and closely aligned coefficient values reported across prior studies support the feasibility and applicability of the proposed reliability modeling approach. Despite these advancements, the absence of a standardized and comprehensive database of degradation parameters remains a major limitation. By identifying critical degradation mechanisms and analyzing widely used modeling parameters, this work highlights existing research gaps and the need for further data-driven refinement of reliability models. Bridging these gaps will improve the accuracy of lifetime prediction, reduce dependence on costly and time-consuming stress testing, and enable more efficient reliability assessments. This study is intended to support researchers in gaining deeper insights into the key failure mechanisms and their reliability modeling by providing standardized lifetime models, associated coefficients, and estimation methodologies. Consequently, this contributes toward accelerated development and deployment of reliable WBG-based PE systems.

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